Pimpri Chinchwad Education Trust's

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## Department of Information Technology

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# AI-Based Urban Climate Resilience Score Using Multimodal Environmental Intelligence

Course Name-Artificial Intelligence and Machine Learning

Formative Assessment I

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# MOTIVATION

**Increasing climate risks:** Urban areas are increasingly experiencing heatwaves, flooding, poor air quality, and loss of green spaces.

**Data is available but fragmented:** Cities collect large amounts of environmental and infrastructure data, but it is often analyzed separately.

**Need to identify vulnerable areas:** A city may have zones that are much more exposed to climate risks than others, which requires zone-level assessment.

**Existing systems are reactive:** Most systems monitor individual conditions rather than predicting future risks and overall resilience.

**Support better decision-making:** A unified resilience score can help authorities prioritize areas that need immediate intervention.

# PROBLEM STATEMENT

Climate change is increasing heatwaves, urban flooding, poor air quality, and loss of green spaces, affecting city residents, public health, infrastructure, and urban sustainability. Although cities collect data through weather stations, air-quality sensors, satellite imagery, and IoT devices, existing systems mainly monitor individual parameters and do not integrate them to provide a comprehensive assessment of climate resilience across different city zones. As a result, vulnerable areas and future climate risks may not be identified effectively, limiting timely and informed urban planning. Therefore, an integrated solution is needed to assess and improve urban climate resilience. The proposed AI-Based Urban Climate Resilience Score will combine climatic, environmental, and infrastructure data using machine learning to evaluate resilience levels of city zones, identify vulnerable areas, predict future climate risks, and recommend suitable adaptation measures such as increasing green cover, improving drainage, and reducing urban heat islands.

# LITERATURE SURVEY

| Paper Title | Publication Details & Year | Dataset Details | Method / Algorithm | Key Findings |
|---|---|---|---|---|
| Urban Climate INTelligence (UCINT): A scalable framework for resilient cities | Molla – Climate Phys. & Atmos. Sci., 2026 | No fixed dataset — framework uses real-time sensing, climate, participatory-governance data | GeoAI; Random Forest, SVM, CNN, LSTM | Scalable multi-hazard framework mapping vulnerability, forecasting hazards & institutional readiness |
| Intra-urban air pollution predictions (low-cost sensors + satellite + GeoAI) | Liang et al. – Environ. Pollution, 2023 | Low-cost sensor network + gap-filled satellite AOD + airborne LiDAR 3D urban form | Multi-scale attention CNN; kriging | 30 m PM2.5 estimates revealing block-level pollution inequality |
| Explainable GeoAI Framework for Flood Susceptibility & Urban Exposure | Tella et al. – Sustainable Development, 2026 | Topo/hydrological/climatic /anthropogenic predictors | RF, Extra Trees, XGBoost, LightGBM, CatBoost, CNN, DNN; SHAP/ALE | Explainable flood susceptibility; CatBoost best (AUC 0.956) |

| Paper Title | Publication Details & Year | Dataset Details | Method / Algorithm | Key Findings |
|---|---|---|---|---|
| U-HEAT: Urban Heat Vulnerability Analysis (ML + Remote Sensing) | Li et al. – Remote Sensing, 2024 | LST, land use, population density, NDVI, NDBI | ML integrated with remote sensing; active learning | Retrospective & forward-looking heat vulnerability mapping |
| Urban Flood-Risk Assessment: Decision-Making & Machine Learning | Taromideh et al. – Sustainability, 2022 | Topo/hydrological/climatic + socio-economic (Rasht, Iran) | AHP + CART | River distance, drainage & population density dominate; works with limited data |
| Modelling urban heat island effects: global analysis of 216 cities | Kong et al. – Computational Urban Science, 2025 | Climate, economic, population, land-use data, 216 cities (2019–2023) | Support Vector Regression | Global UHI prediction (MAE 0.86 °C) across climate zones |
| Prioritizing environmental determinants of urban heat islands (China) | Hou et al. – Int. J. Applied Earth Obs., 2023 | Multi-source RS: greenery fraction, albedo, morphology, human activity (17 Chinese cities) | Random Forest | Greenery fraction & surface albedo are top UHI determinants |
| Sustainable UHI Mitigation through ML: physical & social determinants | Syeda et al. – Sustainability, 2025 | Imperviousness, NDVI, building area, solar influx + income/age/population across 5 zones | RF, SVM, Gradient Boosting Machine | Zone-specific UHI mitigation using physical + socio-demographic drivers |

# RESEARCH GAP

- Existing machine learning studies address urban climate hazards in isolation, focusing on heat, flood, or air pollution independently rather than an integrated assessment.
- Consequently, no unified, zone-level urban climate resilience score yet combines multiple hazards into a single, explainable metric with recommended adaptation actions, and current systems largely monitor individual parameters such as temperature or pollution.
- Reliability is further limited by scarce and imbalanced event data, and by evaluation that remains geographically concentrated and inconsistently reported, with existing integration frameworks mostly conceptual and unvalidated against real-world multi-hazard outcomes.
- This indicates a need for a class-imbalance-robust, data-efficient, explainable resilience score that integrates environmental, climatic, and urban infrastructure data and is validated across multiple cities against observed multi-hazard outcomes.

# OBJECTIVE

1. To collect and integrate satellite imagery, IoT sensor data, weather data, and air-quality measurements for urban climate analysis.

2. To develop a machine learning model to compute an Urban Climate Resilience Score (0–100) for different city zones.

3. To predict the vulnerability of urban zones to heatwaves, flooding, and poor air quality using the integrated data.

4. To generate explainable, data-driven recommendations for climate adaptation, such as increasing green cover and improving drainage.

# DATASET DETAILS/ANALYSIS

| Dataset | Source | Samples / Coverage | Features | Target / Problem Type |
|---|---|---|---|---|
| Copernicus Sentinel-2 | Copernicus Data Space (ESA) | Global, 10m resolution, 5-day revisit | 13 spectral bands, NDVI, land cover, urban footprint | Land cover/vegetation input – feature engineering |
| OpenAQ | OpenAQ (govt + crowdsourced sensors) | Station-level, hourly readings, variable global coverage | PM2.5, PM10, $NO_2$, $O_3$, $SO_2$, CO | Air-quality sub-score – Regression |
| Global Surface Water Explorer | EC Joint Research Centre / Google | Global, 30m resolution, monthly/yearly, 1984–present | Water occurrence, seasonality, change intensity | Flood exposure input – feature engineering |
| ERA5 Climate Reanalysis | Copernicus Climate Data Store (ECMWF) | Global, 0.25° grid, hourly, 1940–present | Temperature, precipitation, wind, humidity | Climate trend input – time-series regression |

| Dataset | Dataset Relevance | Data Characteristics & Feature Analysis | Limitations | Reference |
|---|---|---|---|---|
| Copernicus Sentinel-2 | Captures vegetation & urban expansion driving heat vulnerability | High-res imagery; cloud gaps; NDVI & built-up index are key predictors, need cloud masking | Cloud cover in monsoon season; large preprocessing overhead | dataspace.copernicus.eu |
| OpenAQ | Directly measures the air-quality dimension of urban resilience | Sparse station density, missing values; PM2.5 & $NO_2$ most predictive, need gap-filling | Uneven station coverage, especially in low-income areas | openaq.org |
| Global Surface Water Explorer | Historical water extent supports flood-risk & exposure mapping | Long time series with some cloud/shadow noise; occurrence & seasonality change are key flood indicators | 30m resolution may miss small water bodies/drains | global-surface-water.appspot.com |
| ERA5 Climate Reanalysis | Long-term climate data needed for hazard trend & heat/flood modelling | Modelled reanalysis data; temperature extremes & precipitation intensity are key predictors, may need downscaling | Coarse ~25km grid vs. satellite resolution | cds.climate.copernicus.eu |

# REFERENCES


[1] Molla, A. (2026). Urban Climate INTelligence (UCINT): A scalable framework for resilient cities. Climate Physics and Atmospheric Science, 1–2, 100001. doi:10.1016/j.cpas.2025.100001

[2] Liang, L., Daniels, J., Bailey, C., Hu, L., Phillips, R., & South, J. (2023). Integrating low-cost sensor monitoring, satellite mapping, and geospatial AI for intra-urban air pollution predictions. Environmental Pollution, 331, 121832. doi:10.1016/j.envpol.2023.121832

[3] Tella, A., Zahidi, I., Fai, C. M., Okolie, C., & Pham, Q. B. (2026). From Prediction to Prevention: An Explainable GeoAI Framework for Flood Susceptibility and Urban Exposure Assessment. Sustainable Development. doi:10.1002/sd.70772

[4] Li, F., Yigitcanlar, T., Nepal, M., Nguyen Thanh, K., & Dur, F. (2024). A Novel Urban Heat Vulnerability Analysis: Integrating Machine Learning and Remote Sensing. Remote Sensing, 16(16), 3032. doi:10.3390/rs16163032

[5] Taromideh, F., Fazloula, R., Choubin, B., Emadi, A., & Berndtsson, R. (2022). Urban Flood-Risk Assessment: Integration of decision-making and machine learning. Sustainability, 14(8), 4483. doi:10.3390/su14084483

[6] Anelli, D., Tajani, F., & Ranieri, R. (2022). Urban resilience against natural disasters: Mapping the risk with an innovative indicators-based assessment approach. Journal of Cleaner Production, 371, 133496. doi:10.1016/j.jclepro.2022.133496

# THANK YOU!